

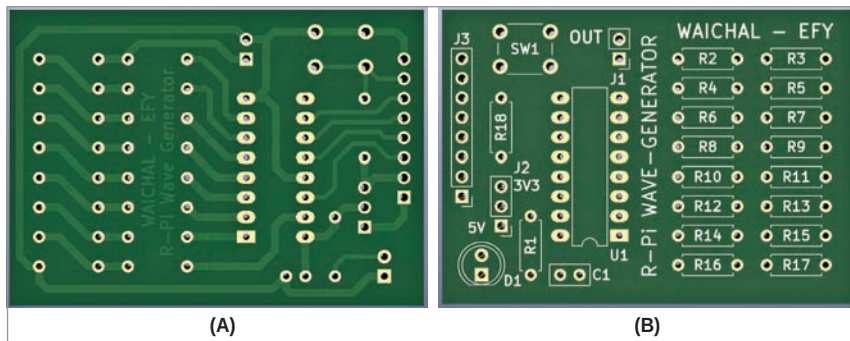


Fig. 3: PCB of the device (A) solder side, (B) components side

span') to be controlled, providing a maximum amplitude of the output waveform as either 3.3V or 5V when the input data is (255)10 or (0xFF) H.

There is also a provision to sense a 'key' input. GPIO pin 18 acts as an input to the Raspberry Pi board and can be used to implement waveform/function selection. This is used only to illustrate the concept of keyboard control of the waveform generator. Users can write their own

software and add additional functions with more key inputs to implement various functionalities, such as frequency control and amplitude control.

EFY note. The DAC operation has been tested with individual software for different waveform outputs. However, it has been found that implementing keyboard control consumes considerable time and may affect the DAC operation and the output waveform update rate.

Connections with Raspberry-Pi GPIO connector

Table 2 shows the interface connections of this waveform generator device with the GPIO connector on the Raspberry Pi board. The connector J3 is used to interface with the Raspberry Pi host computer through

its GPIO pin connector. Testing was performed with the Raspberry Pi board Version 3, which has a 26-pin GPIO connector. Newer versions, such as the Pi 4, have a 40-pin GPIO connector and can also be used, as the GPIO pins are compatible with both versions.

Fig. 3 shows the PCB of the device. The first prototype of the circuit was tested on a breadboard, and later, a single-sided PCB was designed and tested again. Fig. 3A shows the solder side of the PCB, while Fig. 3B shows the components side.

Software code

To run the Python code, install Python 3 (or Python 2, if using an older version) on the Raspberry Pi. If using the latest Raspberry Pi OS, it should come pre-loaded. Various Python codes are provided with the article for generating different waveforms, such as sine wave, square wave, forward ramp, reverse ramp, and triangular wave. Each code can be loaded and executed on the Raspberry Pi computer. Alternatively, users can write their own code to generate arbitrary waveforms, as needed.

To run the code, open the Python file in an IDE and execute

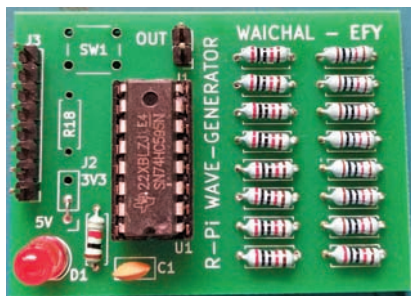


Fig. 4: Completely assembled board for the waveform generator device

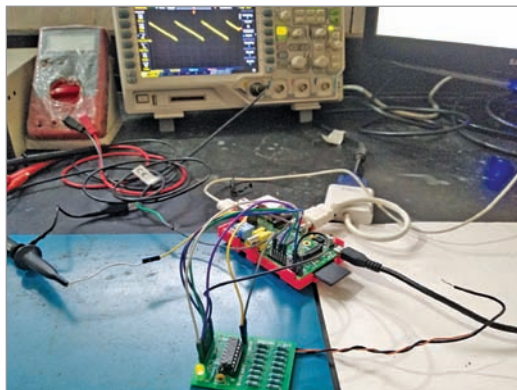


Fig. 5: Testing with the Raspberry Pi board

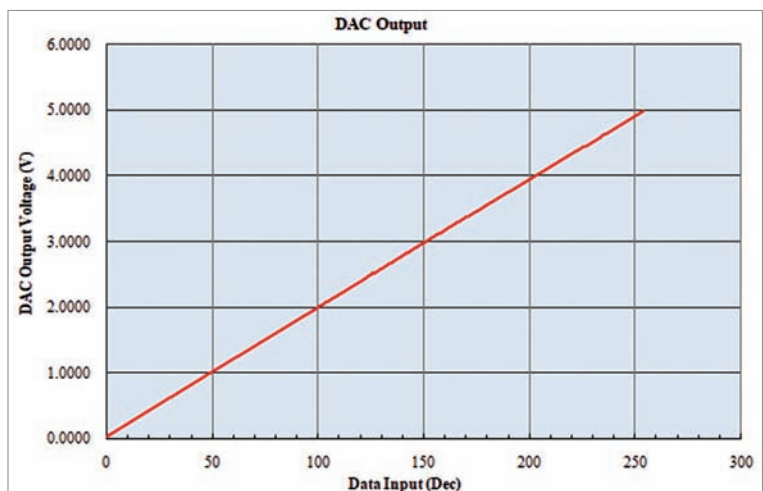


Fig. 6: Linearity of the DAC output for digital data input